INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

RESCUE AND SALVAGING KIT, HYDRAULIC

The General Service Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE

1.1 Scope: This commercial item description covers the Rescue Tool Set (RTS). The RTS is a complete hydraulic rescue tool ensemble composed of commercially available components, including; a diesel engine driven hydraulic power unit, a compressed air driven hydraulic power unit, hydraulic hose reels, plus, spreading, cutting, spreading rams, and combination tools. All tools will operate in all environmental extremes, and under water, from either hydraulic power source, interchangeably.

2.0 SALIENT CHARACTERISTICS

2.1 RTS Description. The RTS described herein is a complete hydraulic rescue tool ensemble composed of a diesel fuel (JP-8) driven hydraulic power unit, with two 100 foot minimum capacity hose reels (each with two 50 foot hose sections), one compressed air driven hydraulic power unit capable of being powered from either a vehicle or a portable compressed air supply, one spreading tool, one cutting tool, three spreading rams, and one aircraft combination tool. The compressed air supply for the hydraulic power unit shall be a one hour self contained breathing apparatus (SCBA) bottle, complete with a manifold, a regulator and a support frame assembly and shall be included in the RTS. All tools must be operable under water and from either the diesel or the compressed air source of hydraulic power with complete interchangeability. This RTS ensemble of components is intended for use in rescue operations involving automotive, structural, and aircraft accidents.

2.2 Design and construction

2.2.1 Environmental and operational requirements. All RTS components shall be fully operational over a temperature range of -20 to +125 degrees F ambient air. RTS tools shall operate while under water and/or after being vertically dropped on concrete from a height of two feet.

Beneficial comments (recommendations, additions, deletions, clarifications, etc.) and any Data that may improve this document should be addressed to: WR-ALC/LEEG, 420 Second Street , suite 100, Robins AFB, GA 31098-1640.

AMSC N/A FSC 4240

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2.2.2 Materials and Components. Materials used for the construction of the RTS and its components shall be the same quality as the manufacturers standard commercial products. Materials shall be free from all defects and imperfections that may affect the serviceability and appearance of the finished product.

2.2.2.1 Fungus Proof Materials. Except where no alternative exists, materials that are nutrients for fungi shall not be used in construction of the RTS. If a fungi nutrient material must be used, such materials shall be treated with a fungicidal agent.

2.2.2.2 Combustible Materials. To the greatest extent possible, non-combustible materials shall be used for construction of the RTS.

2.2.2.3 Hazardous Materials. Materials used in the construction of any RTS component shall have no adverse affect on the health of personnel while being used as part of their normal function. RTS components shall not contain asbestos, lithium, mercury, or radioactive materials.

2.2.2.4 Metals. To the greatest extent possible, lightweight metals shall be used for fabrication of RTS components. Metals used for component fabrication shall be inherently corrosion resistant or shall be plated with a corrosion resistant finish.

2.2.2.5 Fastening Devices. All threaded fasteners shall be installed with a means for preventing loss of torque. Fasteners shall not be swaged, peened or staked. Tapped threads in aluminum shall survive fastener retorting not less than twenty times.

2.2.2.6 Hydraulic Fluid. Hydraulic fluid shall be a synthetic, hydrocarbon base, and shall operate the tools over the ambient temperature range specified herein. All RTS components shall be charged with hydraulic fluid prior to shipping.

2.2.2.7 Hydraulic Quick Disconnect Couplings. Hydraulic couplings for the hoses and tools shall be quick acting, with locking features to prevent accidental uncoupling.

2.2.3 Foreign object damage. All loose metal parts, such as retaining pins and protective caps, shall be secured to the power units, hoses, and the tools so they cannot become separated.

2.2.4 Human Engineering. Human engineering design criteria from the fifth percentile female to the ninety-fifth percentile male shall be applied to RTS operation parameters.

2.2.5 Human Safety. The design of each RTS component shall be such that when the unit is stored, transported, operated, or maintained, in accordance with applicable procedures/precautions, it shall not pose an undue hazard to personnel. Maximum sustained operating noise level shall not exceed 84 dba.

2.2.6 Workmanship. RTS components shall be manufactured in accordance with the requirements specified herein and in accordance with the best commercial practices of the respective industry.

2.2.7 Order of Precedence. If a conflict exists between the text of this document and any

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reference cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws or regulations.

2.3 Diesel Engine Driven Hydraulic Power Unit with Hose Reels. The diesel engine shall operate on JP-8 (MIL-DTL-83133E) without detrimental effect. The power unit shall be equipped with two live hose reels, one on each end of the power unit. Each hose reel shall be equipped with 100 feet of hose, made up of two 50-foot hose sections. Hose reels shall rewind manually. Pump and engine controls shall be mounted on the same side that the hoses deploy. The power unit shall store in a space not to exceed 24 inches high, 40 inches long and 26 inches deep. The power Unit shall be portable and shall be enclosed in a roll cage type frame to protect the components from damage. Hand holds shall be provided on the roll cage frame. The power unit shall not slide or roll over when operating in any orientation on a dry hard surface, inclined at an angle of 15 degrees above the horizontal. When specified (see 6.2), provide an additional complete diesel engine driven hydraulic power unit, with hose reels and hose.

2.4 Compressed Air Hydraulic Power Unit with Air Supply. The compressed air hydraulic power unit is intended for use with any tool while in explosive environments. The compressed air power unit shall connect to the air supply hoses with quick-disconnect fittings. The fully charged weight of the air powered hydraulic pump shall not exceed 35 pounds. A portable air supply consisting of one 60-minute firefighter SCBA air cylinder (4500 psig) with a manifold, regulator, and support frame assembly shall also be provided. Compressed air supply hoses shall be not less than eight feet long. Hydraulic hoses between the air powered hydraulic power unit and the tools shall be not less than 25 feet long. When specified (see 6.2), provide an additional complete compressed air driven hydraulic power unit.

2.5 Spreading Tool. The spreading tool shall be fully operable by one person and shall include removable multi-purpose tips for gripping, spreading, or piercing. Tool tips shall be replaceable without the use of additional tools. The spreader shall not exceed 50 pounds in weight with any set of tips attached. The spreader arms shall be designed so that pulling chain shackles can be attached. The spreader shall be provided with two six-foot chains, each having a removable hook at one end and a removable shackle at the other. Spreading distance at the tips shall be not less than 27 inches. The spreader shall exert a minimum spreading force of 20,000 pounds and a minimum pulling force of 14,000 pounds. The spreader shall pierce and open a 25 by 25-inch access hole through a 1/8 inch thick 7075-T6 aluminum plate. The spreader shall hold the forcing load for a minimum of five minutes when hydraulic pressure is lost or if the deadman control in released. When specified (see 6.2), provide an additional spreading tool.

2.6 Cutting Tool. The cuffing tool shall be fully operable by one person and shall not exceed 40 pounds in weight. The cutting blades shall be of a double acting design with a blade opening not be less than 4.5 inches at the tips. The cutter shall be capable of shearing a 1-¼ inch diameter 1080 steel bar. When specified (see 6.2), provide an additional cutting tool.

2.7 Spreading rams. The spreading ram set shall include short, medium, and long extension rams. Each ram shall be of a double acting design with a minimum spreading force of not less than 15,000 pounds. The extension capability of the three ram shall overlap such that the rams can be used in succession. The ram set shall have a total extension capability of at least 80

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inches. Telescoping rams or ram extensions exceeding three inches long shall not be permitted. Rams must be able to hold the forcing load for a minimum of five minutes after hydraulic pressure is lost or if the deadman control is released. When specified (see 6.2), provide an additional spreading tool.

2.8 Aircraft Combination Tool. The aircraft combination tool (ACT) shall require only one person to operate and shall provide spreading, pulling, cuffing, and piercing capabilities, with a maximum weight not exceeding 35 pounds. The spreader shall open to not less than 12 inches, with a force of not less than 20,000 pounds. The spreader shall have a pulling force of not less than 14,000 pounds, when using a chain kit. The forcing loads shall be held for a minimum of five minutes after a loss of primary pressure. The cutter jaws shall open not less than seven inches and produce not less than 60,000 pounds of cutting force at the blade notch. The cutter shall cleanly shear the braided stainless steel reinforced hose of an aircraft ejection seat (approximately 5/8 inches in diameter) cleanly, without leaving any uncut braiding. The ACT shall cut through 2 by 2 by ¼ inch wall 1022 square steel tubing and shall shear .090 inch thick 2024-T6 aluminum sheet for 9 minutes on one fully charged SCBA compressed air cylinder. The piercing tool tips shall facilitate manual penetration of aircraft composite/metal skins, to gain a gripping edge or a shearing point, and shall be capable of piercing and opening a 10 by 10 inch hole in .060 inch 7075-T6 aluminum plate without damage to the tips or the tool jaws. Two six-foot chains, each having a removable hook at one end and a removable shackle at the other end shall be provided with the combination tool. Closed dimensions of the ACT shall not exceed 36 inches long by 12 inches square. When specified (see 6.2), provide an additional aircraft combination tool.

2.9 Hydraulic Hoses. The RTS shall include two spare 50-foot hose sections for the diesel engine driven hydraulic power unit. When specified (see 6.2), provide an additional set of two spare 50 foot hose assemblies. The RTS shall include one spare 25-foot hose section for the compressed air driven power unit. When specified (see 6.2), provide one additional 25 foot hose assembly. All hydraulic hoses shall be non-conductive, with permanently attached end fittings. Hydraulic hose assemblies shall withstand a burst pressure of four times the working pressure and a proof pressure of two times the maximum operating pressure.

2.10 Tool Considerations.

2.10.1 Carrying Handles and Deadman Controls. All RTS tools shall include carrying handles and shall be operated through deadman type controls. The operator must be able to rotate the tool a minimum of 180 degrees and operate the deadman control while wearing fire-fighter type gloves. Carrying handles shall be located on the tools to provide for balance during operation. Deadman controls shall not allow entry of debris that could impair tool operation. The deadman control shall be marked to indicate open and close actions and shall automatically return to the neutral position when released. Carrying handles and deadman controls shall be oriented on the centerline of the tool to permit either hand operation. Deadman controls shall be located behind carrying handles.

2.10.2 Hose Connections. Hydraulic hose connections shall be located so they will not interfere with the operation of the deadman control.

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2.10.3 Tool Attachments. Tool tips shall be secured to the tool to prevent loss or during operation.

3.0 REGULATORY REQUIREMENTS.

3.1 Recovered Material. The offeror/contractor is encouraged to use recovered materials to the maximum extend practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR)

4.0 PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformation: The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 First Unit Testing. All inspections and testing shall be performed as stated herein. Inspections and testing shall be observed and approved by Government representatives.

4.3 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection and test requirements. The contractor may utilize his own facilities or any commercial facility to perform the inspections and tests. The Government, at its own expense, reserves the right to repeat any inspection or test requirement. Failure of any RTS component to meet any of the performance requirement specified herein shall be cause for rejection. The Government may choose to have trained rescue personnel present for certain component tests. The contractor shall offer these Governments rescue personnel the opportunity to operate any RTS component.

4.3.1 Inspection and Tests Procedures and Reports. The contractor shall submit an inspection and test procedure for all elements of the RTS, a minimum of 30 days prior to the first inspection or test, for review by the procuring activity. Prior to the delivery of the first RTS, the contractor shall submit a summary report of Government approval of the individual inspections and tests.

4.4 Inspections and Tests. Inspections and tests shall demonstrate that the RTS will operate under the environmental and emergency conditions cited herein.

4.4.1 Physical Inspections. All RTS components shall be physically inspected to assure compliance with the respective requirements of this document.

4.4.2 Performance Tests. All RTS components shall be performance tested to assure compliance with the requirements specified herein. The performance of the spreader, cutter, and spreading rams, shall be measured using the Diesel Engine Driven Hydraulic Power Unit. The performance of the Aircraft Combination Tool shall be measured using the Compressed Air Hydraulic Power Unit. During testing, no saws, drills, or torches shall be used to assist the RTS.

4.4.2.1 Spreader Test. The spreader tool test shall include;

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a. 20 piercings of 1/8 inch thick 7075-T6 aluminum followed by spreading a 25 x 25 inch hole after every other piercing.

b. Measuring a minimum 20,000 pound spreading force. The spreading force shall hold for a minimum of five minutes with deadman control released and with hydraulic power disconnected.

c. Measuring a minimum 14,000 pound pulling force. The pulling force shall hold for a minimum of five minutes with deadman control released and with hydraulic power disconnected.

4.4.2.2 Cutter Test. The cutter tool test shall include;

a. Measuring a minimum 60,000 pound cutting force at the tool jaw notch.

b. Making 20 cuts of a 1 1/4 inch diameter, 1080 steel bar.

4.4.2.3 Ram Test. The spreading ram test shall include;

a. Measuring a minimum 15,000 pound spreading force for each size ram. The spreading force shall hold for a minimum of five minutes with deadman control released and with hydraulic power disconnected.

4.4.2.4 Aircraft Combination Tool. The aircraft combination tool test shall include;

a. Measuring a minimum 20,000 pound spreading force. The spreading force shall hold for a minimum of five minutes with deadman control released and with hydraulic power disconnected.

b. Measuring a minimum 14,000 pound pulling force. The pulling force shall hold for a minimum of five minutes with deadman control released and with hydraulic power disconnected.

c. Measuring a minimum 60,000 pound cutting force at the tool jaw notch.

d. Making 20 cuts of a braided stainless steel sheathed 5/8-inch hose.

e. Making 20 cuts of 2 by 2 by ¼ inch wall 1022 square steel tubing.

f. Making 20 piercings of .060 inch thick 7075-T6 aluminum and spreading a 10 by 10 inch hole after every other piercing.

g. Making five continuous sheerings of .090 inch thick 2024-T6 aluminum. Each shearing shall be sustained for a minimum of nine minutes and require no more than one fully charged one-hour SCBA cylinder.

4.4.2.5 Drop and Underwater Performance Tests. Each RTS tool shall be dropped on concrete from a vertical distance of 2 feet. Each tool shall then accomplish one cycle of the respective

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tool performance tests. Each tool shall then accomplish one cycle of the respective tool performance tests while submerged in salt water.

4.4.2.6 Environmental Performance Tests. Each tool shall be cold soaked at -20 degrees F. for a minimum of 15 hours followed by accomplishing one cycle of the individual tool performance tests at the cold soak temperature.

5.0 PACKAGING.

5.1 Packaging. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

5.2 Identification Plate. A permanently marked identification plate shall be attached to each RTS component containing the manufacturer's name, model number and serial number.

6.0 NOTES.

6.1 Intended Use. The Rescue Tool Set will be stored/carried on Rescue trucks and will be operated by rescue personnel in emergency situations.

6.2 Ordering Data. Procurement documents should specify the following:

a. When an additional diesel engine driven hydraulic power unit is required (see 2.3).

b. When an additional compressed air driven hydraulic power unit is required (see 2.4).

c. When an additional spreader tool is required (see 2.5).

d. When an additional cutter tool is required (see 2.6).

e. When an additional spreading ram set is required (see 2.7).

f. When an additional aircraft rescue tool is required (see 2.8).

g. When an additional set of two 50-foot hydraulic hoses is required (see 2.9).

h. When an additional 25 foot hydraulic hose assembly is required (see 2.9)

6.3 Commercial Products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurements to only the manufacturers shown.)

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MFR'S CAGE	MFR'S name and address
0MVF1	HURST Jaws of Life 711N Post Road Shelby, NC 28150 TEL: 800-537-2659
8Z488	AMKUS, Inc 2700 Wisconsin Avenue, Downers Grove, IL 60515 TEL: 630-515-1800

NSN 4240-01-248-8551

MILITARY INTERSET:

Custodian:
Air Force - 99

Preparing Activity:
Air Force – 84

Agent Activity:
Air Force – 99

(Project No. 4240-0697)

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